

Arduino Calculator Structure

Console Application

Headers

- The headers include standard and windows specific libraries that help with file handling, string manipulation, and interacting with the Windows API – specifically for COM port communication

Functions

- StringToWString converts a small string into a longer string for Windows API like CreateFileW
- GetParity is used to see errors in sent and received data. In serial communication, the parity bit is used to see if the listening device is receiving the data correctly.
- The toupper(p) is used to convert a smaller case letter to Uppercase.
- The GetStopBits is used to get the desired end bits to mark the end of a data transmission packet.

Main

- // select COM port: The user selects the appropriate communication port from the list (COM1 to COM5).
- // Baud rate: The user selects the correct baud rate according to the board's settings.
- // Parity selection: The user selects the parity bit that the board has been set to.
- // Stop bits: The user selects the stop bits the board has been set to.
- // Open COM port: Where the magic happens, indicating if you have good communication or not! CreateFileW opens the selected COM port.
- // Configure serial: serial communication is configured with DCB structure and SetCommState.
- // Set timeouts: sets the communication timeouts, longer time can cause programs to become unresponsive or hanging.
- // Main Loop: The user's input is sent to the Arduino via WriteFile with std:getline.
- The Arduino's response is checked with ReadFile, and then displays the response on the console, and logs it in log.txt.

Arduino

Libraries:

- LiquidCrystal_I2C.h: Library used for controlling an LCD screen via the I2C protocol.
- Wire.h: This is the standard library for I2C communication.

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Global variables

- The LCD screen is initialized with address 0x27 and size 16x2. Two LED pins (grnLED for green, redLED for red) are used for feedback on the communication status.
- The home screen time duration is managed by prevhometime and displayDuration.

Setup

- The LCD screen and LEDs are initialised.
- Serial communication starts at 4800 baud.
- The home screen is displayed with HomeScr().

Loop

- The code here waits for serial input and then processes it.
- If the calculation does not start with "CALC", it shows an error message.
- Parsing: It extracts the two arguments (values) and the operator from the expression.
- The calculation is performed depending on the operator, and handles errors like division by zero.
- After the calculation has been performed, the result or error message is displayed on the LCD screen.
- Next, the result is sent back to the computer over serial.
- The red LED blinks if an error occurs and green LED blinks for successful communication.
- Home Screen: After 5 seconds of inactivity, the LCD returns to the home screen.

Instructions for program

Console

- The Visual Studio solution can be found in the main Arduino calculator file. Opening this will open up Visual studio.
- The program can be started by entering Ctrl and F5. Or by alternatively opening the x64 and thereafter the debug file to find the application.

Arduino

- The Arduino code can be found in the "Arduino Code" folder. The code needs to be sent to the Arduino.

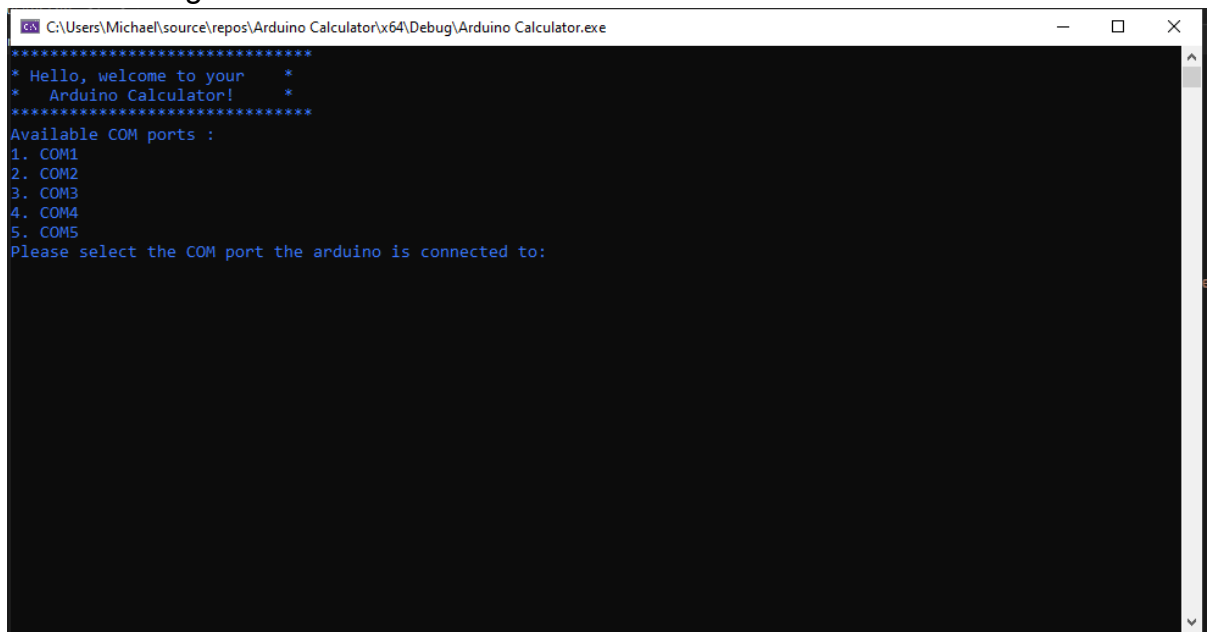
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- After the Arduino has been programmed, the LCD, LED's and serial to USB device may be connected.
- The Arduino would also need to be powered on, by a dc jack, or alternatively a usb power supply.
- The serial unit may be connected to the computer.

Instructions for running program

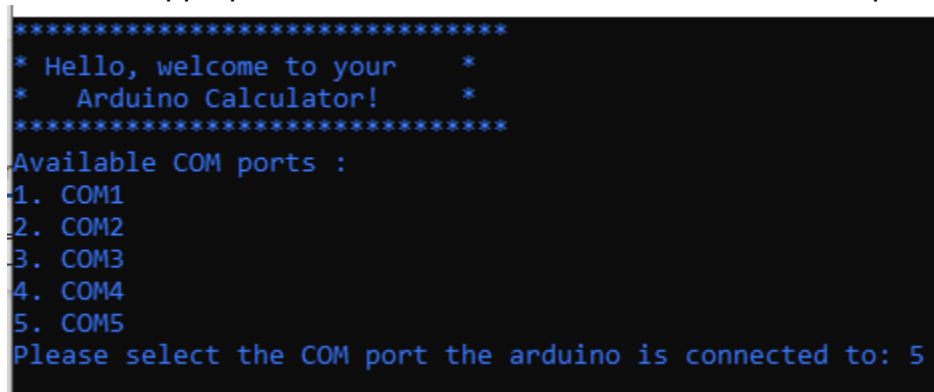
Starting application

- The application can be started by entering Ctrl + F5.
- The following screen will be shown:



```
*****
* Hello, welcome to your *
* Arduino Calculator! *
*****
Available COM ports :
1. COM1
2. COM2
3. COM3
4. COM4
5. COM5
Please select the COM port the arduino is connected to:
```

- Enter the appropriate number the board is connected to and press enter:



```
*****
* Hello, welcome to your *
* Arduino Calculator! *
*****
Available COM ports :
1. COM1
2. COM2
3. COM3
4. COM4
5. COM5
Please select the COM port the arduino is connected to: 5
```

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- Enter the baudrate your Arduino is set to, in our case the Arduino is set to 4800:

```
*****
* Hello, welcome to your      *
*   Arduino Calculator!       *
*****
Available COM ports :
1. COM1
2. COM2
3. COM3
4. COM4
5. COM5
Please select the COM port the arduino is connected to: 5
Please enter the baud rate (default 9600): 4800
```

- Enter the parity bit, ours is set to default, so we can omit this, and press enter:

```
*****
* Hello, welcome to your      *
*   Arduino Calculator!       *
*****
Available COM ports :
1. COM1
2. COM2
3. COM3
4. COM4
5. COM5
Please select the COM port the arduino is connected to: 5
Please enter the baud rate (default 9600): 4800
Please enter the parity bit [N=none, E=even, O=odd] (default N):
```

- Enter the stop bit/s, again our Arduino is set to the default, so we can press enter:

```
*****
* Hello, welcome to your      *
*   Arduino Calculator!       *
*****
Available COM ports :
1. COM1
2. COM2
3. COM3
4. COM4
5. COM5
Please select the COM port the arduino is connected to: 5
Please enter the baud rate (default 9600): 4800
Please enter the parity bit [N=none, E=even, O=odd] (default N):
Lastly, please enter the stop bits [1, 2] (default 1):
```

Arduino Calculator Structure

- After successful connection, you will see the following screen:

```
Lastly, please enter the stop bits [1, 2] (default 1):  
Connected to COM5 @ 4800 baud, now the magic happens!  
Please enter CALC followed by your values you want to be calculated  
For example: CALC 15 x 25, etc.  
>
```

- If not a successful connection, you will see this screen:

```
*****  
* Hello, welcome to your *  
* Arduino Calculator! *  
*****  
Available COM ports :  
1. COM1  
2. COM2  
3. COM3  
4. COM4  
5. COM5  
Please select the COM port the arduino is connected to: 4  
Please enter the baud rate (default 9600): 4800  
Please enter the parity bit [N=none, E=even, O=odd] (default N):  
Lastly, please enter the stop bits [1, 2] (default 1):  
Oh no! We have an error opening the COM port COM4  
  
C:\Users\Michael\source\repos\Arduino Calculator\x64\Debug\Arduino Calculator.exe (process 7112) exited with code 1 (0x1).  
Press any key to close this window . . .
```

- If the wrong baudrate has been selected, you will find this, in which case the program needs to be restarted and right baudrate needs to be selected:

```
Lastly, please enter the stop bits [1, 2] (default 1):  
Connected to COM5 @ 9600 baud, now the magic happens!  
Please enter CALC followed by your values you want to be calculated  
For example: CALC 15 x 25, etc.  
> CALC 1+2  
↑f>
```

- After successful connection, commands can be sent in the form: CALC 1 + 56, etc:

```
Connected to COM5 @ 4800 baud, now the magic happens!  
Please enter CALC followed by your values you want to be calculated  
For example: CALC 15 x 25, etc.  
> CALC 1 + 56  
RAW: [CALC 1 + 56]  
RESULT: 57.0000  
>
```

Arduino Calculator Structure

- This will be shown successively on the lcd screen:

